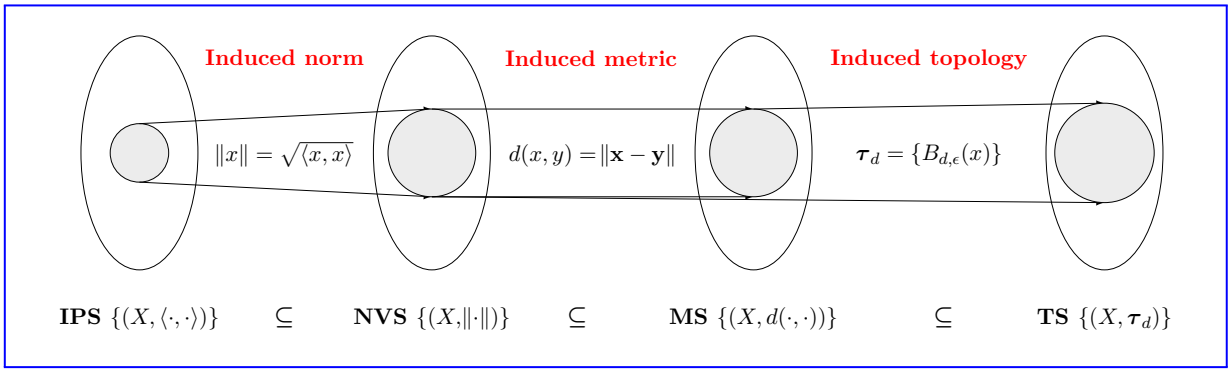


Figure 1: Induced Norm, Metric, and Topology

| | | | |
|---|---|---|---|
| **IPS** | $--$ | Inner Product Spaces | $(X, \langle \mathbf{x}, \mathbf{y}\rangle)$ |
| **NVS** | $--$ | Normed Vector Spaces | $(X, \|\mathbf{x}\|) = (X, \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle})$ |
| **MS** | $--$ | Metric Spaces | $(X, d(\mathbf{x}, \mathbf{y})) = (X, \|\mathbf{x} - \mathbf{y}\|) = (X, \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y}\rangle})$ |
| **TS** | $--$ | Topological Spaces | $(X, \boldsymbol{\tau}_d) = (X, \{B_{d,\epsilon}(\mathbf{u})\})$, where $U \subseteq X$ and $(\forall \mathbf{u} \in U,\ \exists \epsilon > 0) : B_{d,\epsilon}(\mathbf{u}) \subseteq U$ |

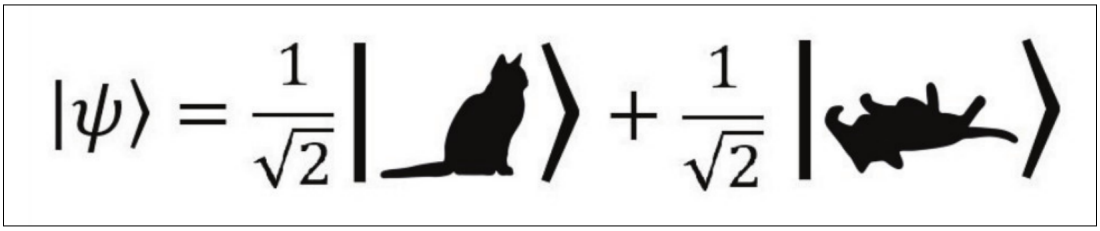


Figure 2: Quantum superposition and Schrödinger's cat

$n=\langle\ ,\ \rangle$
$G(\ )$
$\ \ \ \

0.6249999121401877

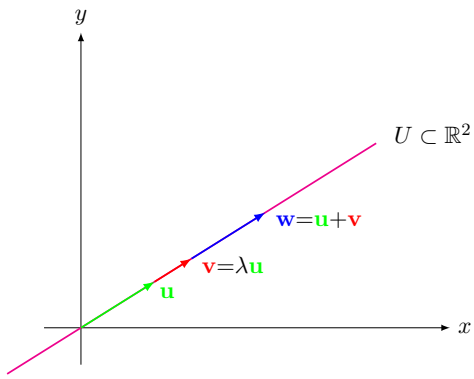


Figure 5: The line $U$ is a linear subspace of $\mathbb{R}^2$

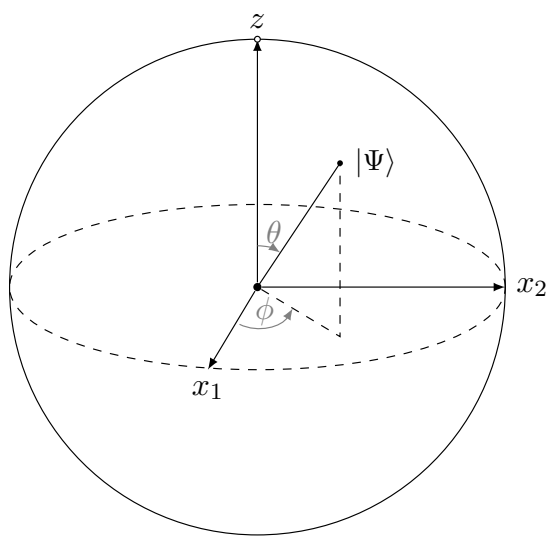


Figure 6: The Bloch Sphere

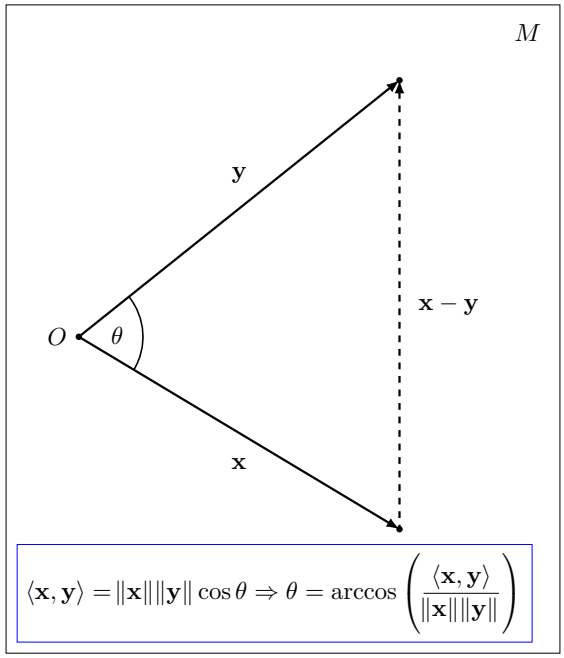


Figure 7: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

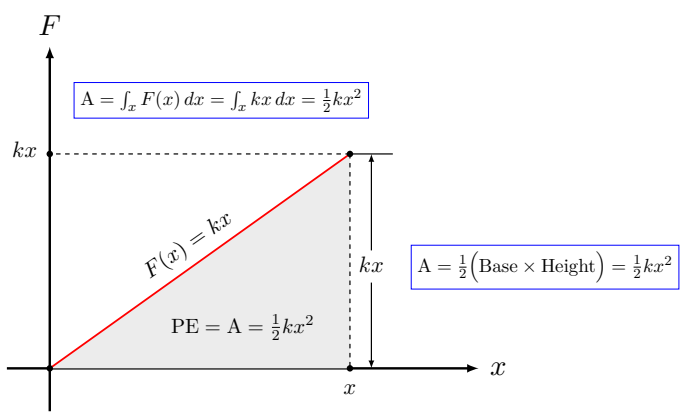


Figure 8: Potential Energy of a Simple Harmonic Oscillator

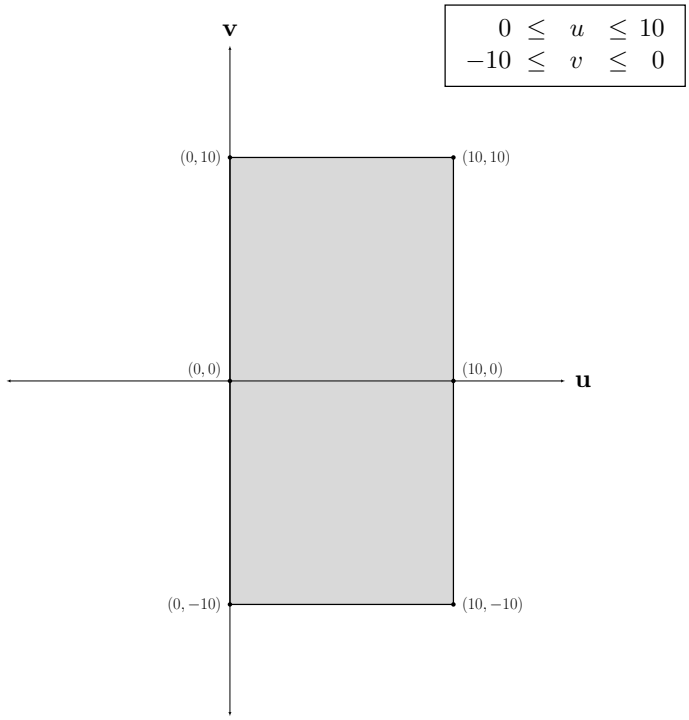


Figure 9: The transformed parallelogram is a rectangle in the $uv$-plane

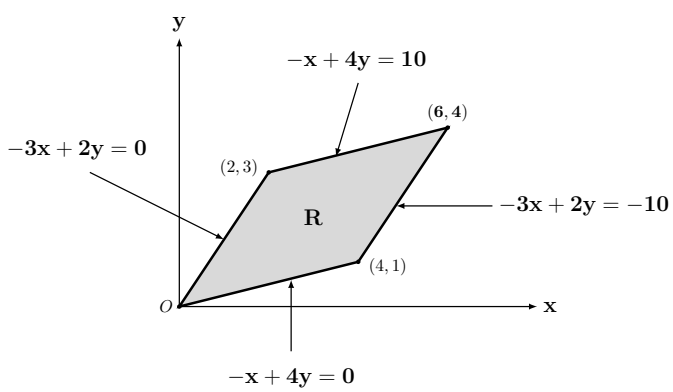


Figure 10: Equations of the Sides of the Parallelogram

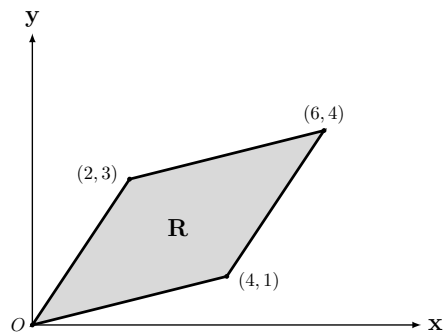


Figure 11: Parallelogram in the xy-plane

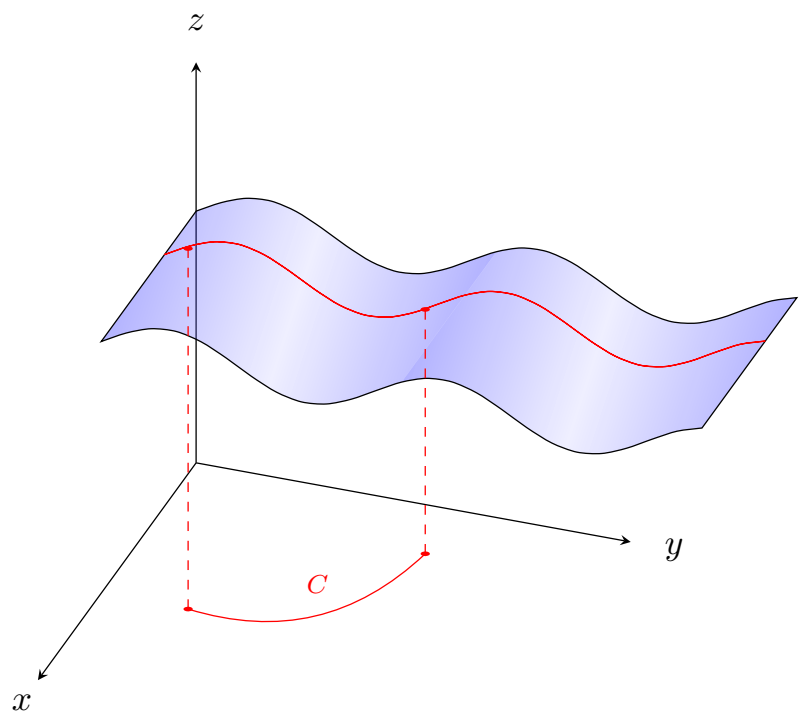


Figure 12: 3D

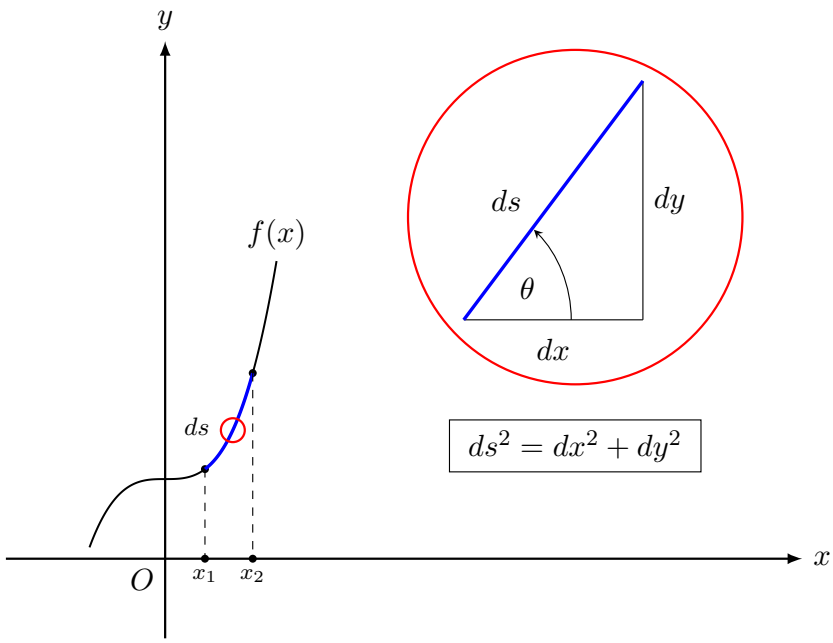


Figure 13: $f(x)$, $ds$ and the Pythagorean Theorem

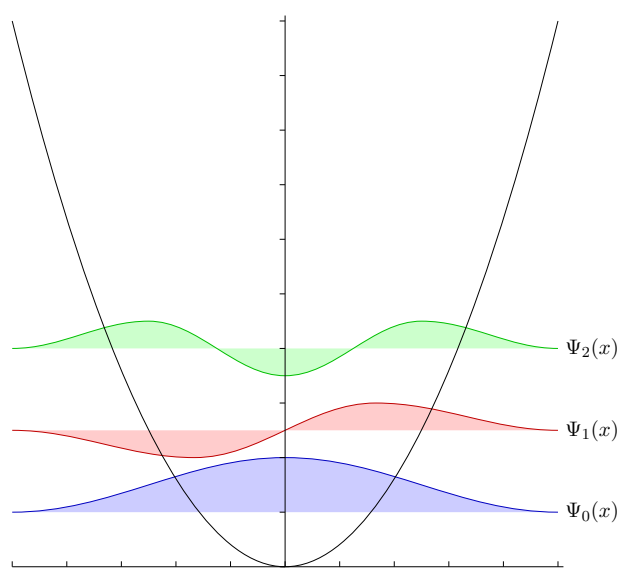


Figure 14: Wave Functions

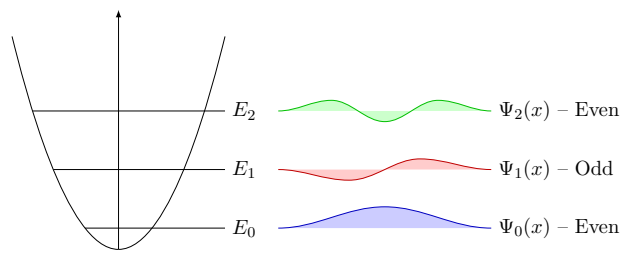


Figure 15: Wave Functions

| | | | | | | |
|---|---|---|---|---|---|---|
| $9^2$ | $=$ | $81$ | $\longrightarrow$ | $8+1$ | $=$ | $9$ |
| $45^2$ | $=$ | $2025$ | $\longrightarrow$ | $20+25$ | $=$ | $45$ |
| $703^2$ | $=$ | $494209$ | $\longrightarrow$ | $494+209$ | $=$ | $703$ |
| $7777^2$ | $=$ | $60481729$ | $\longrightarrow$ | $6048+1729$ | $=$ | $7777$ |
| $857143^2$ | $=$ | $734694122449$ | $\longrightarrow$ | $734694+122449$ | $=$ | $857143$ |

Figure 16: A Few Example Kaprekar Numbers

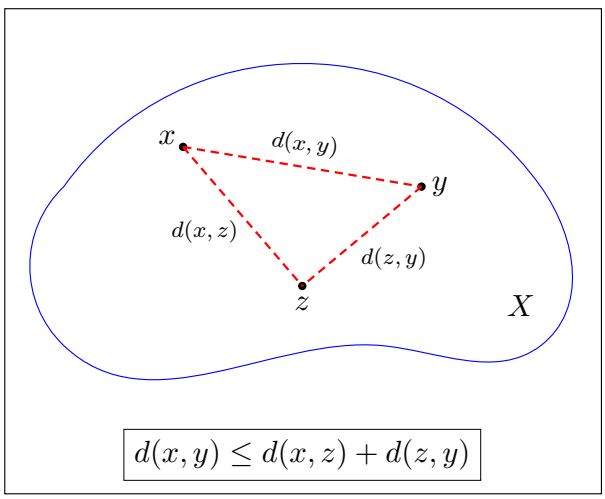


Figure 17: The Triangle Inequality

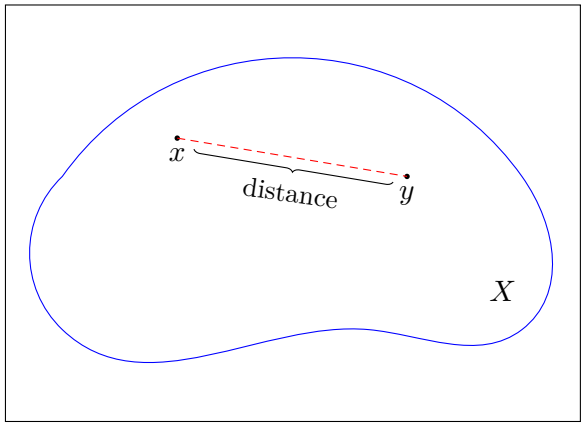


Figure 18: $X, x$ and $y$

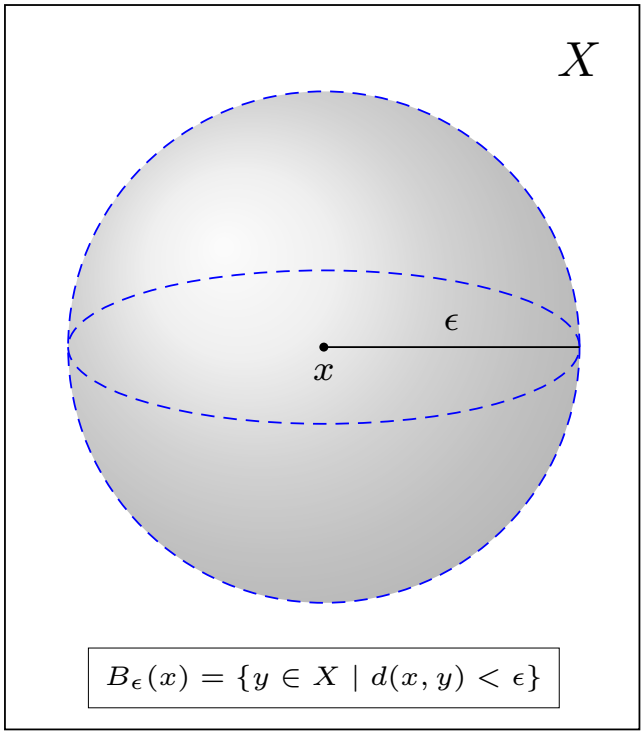


Figure 19: $B_\epsilon(x)$ is an open epsilon ball centered at $x$ with radius $\epsilon$

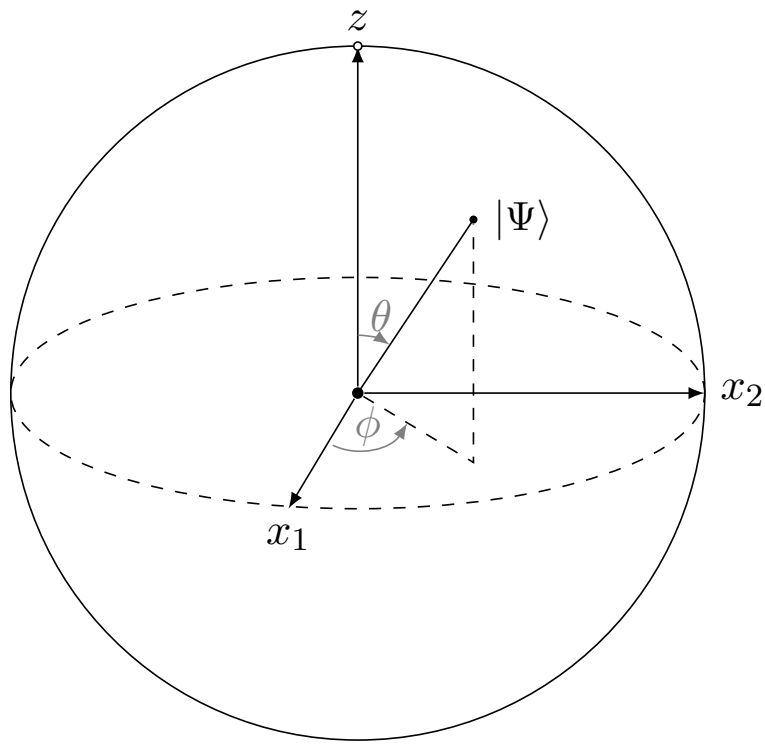


Figure 20: The Bloch Sphere

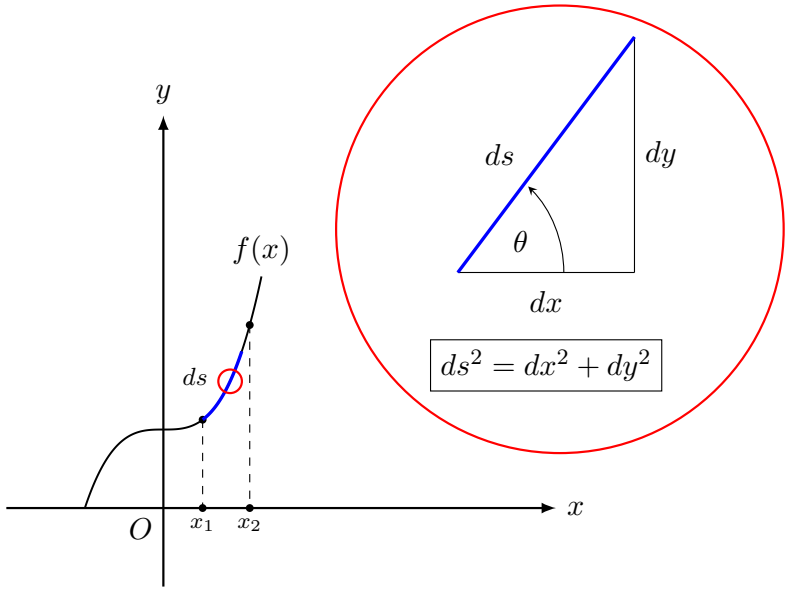


Figure 21: $f(x)$, $ds$ and the Pythagorean Theorem

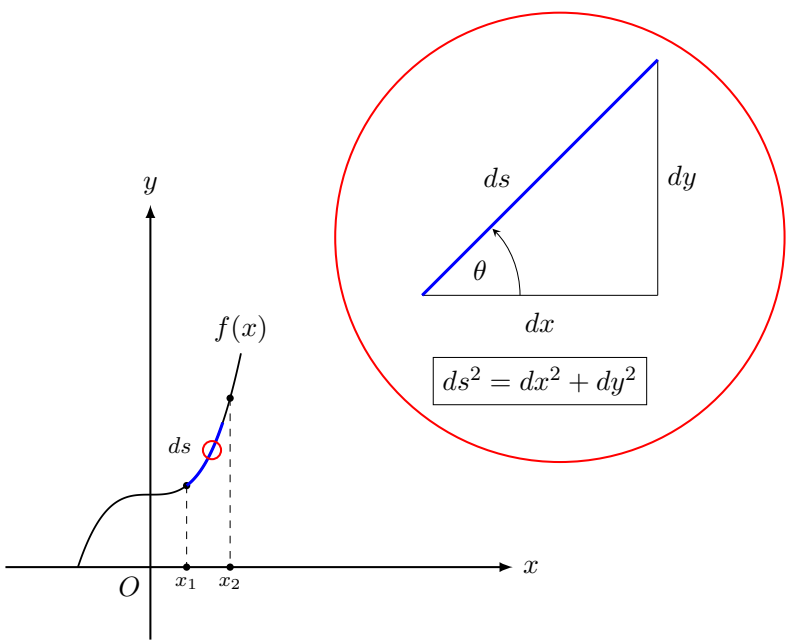


Figure 22: $f(x)$, $ds$ and the Pythagorean Theorem